A Hybrid Framework for Recommendation via LLM Semantic Anchoring and Static Graph Filtering

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Abstract. We combine a frozen semantic item graph with item-side contrastive alignment, using RLMRec's released semantic embeddings. Over five paired seeds, our framework improves test Recall@20 over full RLMRec by 0.00705 on Amazon-book and 0.00474 on Steam, and loses by 0.00281 on Yelp. The direction is consistent across all five seeds on each dataset. Steam inherits Amazon's model hyperparameters, with a different batch size and stopping budget. Exploratory controls show that removing alignment can retain a validation advantage over selected baselines, while permuting item labels on the graph reduces performance. Those controls differ in protocol and do not isolate the complete mechanism of the main-test advantage. We report the Yelp loss explicitly; a recorded semantic-agreement prediction failed on Steam, leaving the cross-dataset boundary unresolved.

Keywords: Recommendation · Language models · Graph neural networks · Semantic item graphs

1 Introduction

RLMRec aligns collaborative representations with semantic vectors obtained from generated user and item profiles [1]. The same item vectors can also define a similarity graph that supplies information during propagation, following the frozen-graph idea in FREEDOM [2]. These uses are complementary: one changes the training objective and the other changes the representation computed by the model.

Our framework combines a frozen cosine nearest-neighbour item graph with an item-side alignment loss and a LightGCN collaborative branch. We compare it with full and item-only RLMRec-Con on Amazon-book, Yelp and Steam using the same released semantic inputs. Main model comparisons share a dataset's evaluation and stopping protocol, but the recipes differ in several architectural and objective choices. Their performance difference cannot be attributed solely to the addition of an item graph.

The result is a consistent but conditional advantage: the hybrid framework beats full RLMRec on Amazon and Steam and trails it on Yelp. Steam uses model hyperparameters transferred from Amazon without a Steam-specific sweep; its training batch and patience differ from Amazon's. Supporting experiments remove the alignment loss or permute graph labels to examine how the recipe uses

item semantics. We distinguish these exploratory controls from the five-seed test comparisons. They motivate further investigation but do not explain why the recipe loses to RLMRec on Yelp.

2 Method

Backbone and objective. Let B training triples (u, i, j) contain an observed item i and a sampled negative j. With collaborative embeddings h and $\hat{y}_{ui} = h_u^\top h_i$, the averaged BPR loss is

$$\overline{\mathcal{L}}_{\text{BPR}} = \frac{1}{B} \sum_{(u,i,j)} \text{softplus}(\hat{y}_{uj} - \hat{y}_{ui}). \tag{1}$$

RLMRec-Con adds batch-averaged cosine InfoNCE terms on users and on positive and negative items, through a shared two-layer projector that maps frozen 1536-dimensional semantic vectors into the collaborative dimension. The full objective is $\mathcal{L}_{\text{full}} = \overline{\mathcal{L}}_{\text{BPR}} + \alpha(\overline{\mathcal{L}}_u + \overline{\mathcal{L}}_{i^+} + \overline{\mathcal{L}}_{i^-}) + \beta\|\Theta\|_2^2$, and dropping $\overline{\mathcal{L}}_u$ gives the item-only variant.

Frozen semantic item graph. The hybrid framework builds a cosine top-k graph S over the item semantic vectors, excluding self-links, takes the binary symmetric union and normalises it as $D^{-1/2}SD^{-1/2}$. Item ID embeddings propagate once over this fixed graph and their unit-normalised output is added to the collaborative item embeddings. The collaborative branch uses mean layer readout and samples 80% of training edges per epoch with weights proportional to $1/\sqrt{d_u d_i}$; evaluation uses the complete training graph. Alignment, when enabled, applies only to positive items' fused representations. Setting $\alpha = 0$ removes the alignment term entirely and leaves the graph as the only semantic input.

Relation to prior recipes. Semantic and learned item graphs are established: LATTICE learns item relations from side features [5], FREEDOM freezes them and samples the collaborative graph [2], and SAGERec constructs an LLM-derived item graph for sequential recommendation [6]. AlphaRec shows that item language representations alone support strong recommendation [4]. Our contribution is not the idea of a semantic item graph but the comparison of complete recipes against profile alignment on shared inputs and a matched main-comparison protocol within each dataset.

We audited our implementation against the released FREEDOM code and record the differences rather than claiming a reproduction. Ours excludes self-links where upstream leaves them eligible, symmetrises before normalising where upstream applies a directed row normalisation, unit-normalises the structural output before addition, prunes 20% of edges where upstream drops 80–90%, and replaces upstream's feature-level BPR terms with an InfoNCE term on a single frozen semantic modality. The resulting framework is FREEDOM-inspired, not FREEDOM with different features, and our results do not speak to the performance of the original model.

3 Experimental Design

Data and protocol. We use RLMRec's released interaction matrices and semantic embeddings for all three benchmarks: Amazon-book (11,000 users, 9,332 items, 120,464 training interactions), Yelp (11,091 / 11,010 / 166,620) and Steam (23,310 / 5,237 / 316,190). Splits are approximately 60/20/20 and we verified that the three matrices share no nonzero pair on any dataset, and that every semantic array has the expected entity count and 1,536 finite dimensions. Collaborative adjacency uses training data only. All-ranking evaluation masks training positives; validation positives remain eligible test candidates. Evaluation metrics are macro-averaged over users with positives in the evaluated split.

We cannot verify a complete split-filtered generation history for the released profiles. The upstream description cites GPT-3.5-turbo and text-embedding-ada-002, but published examples are not generation logs. Our results are conditional on these fixed inputs; disjoint interaction splits do not certify the absence of contamination upstream.

Configurations. All compared models use 32-dimensional collaborative embeddings, three collaborative layers, Adam at learning rate 10^{-3} and no optimiser weight decay. RLMRec variants use $\alpha = 0.01$, temperature 0.2, keep rate 0.8 and $\beta = 10^{-7}$. Batch size is 1024 on Amazon and 4096 on Yelp and Steam, fixed within each dataset; cross-dataset behaviour is therefore not an isolated effect of dataset identity. The hybrid framework uses $\alpha = 0.03, k = 10$ on Amazon and $\alpha = 0.1, k = 20$ on Yelp, with one item-graph layer and pruning ratio 0.2.

Steam inherits Amazon's model hyperparameters, including α, k, embedding dimension, layer counts, temperature, regularization and pruning. Its batch size and stopping budget differ. The recorded study transferred this recipe without a Steam-specific model hyperparameter sweep; each confirmation run evaluates the test set once at its best validation checkpoint. Earlier Steam development runs and historical tests remain in the artifact. This is retrospective transfer evidence, not a claim of an untouched test set or a fully preregistered study.

Stopping and checkpoint selection. Validation Recall@20 is evaluated every three epochs. Patience is five non-improving checks for the Amazon and Yelp main comparisons and 25 for Steam, matched across the four models within each dataset. The best checkpoint is restored before one final test evaluation per run. All 60 manifest runs ended by early stopping. The paired seeds are 1, 7, 13, 42 and 2024.

Separate Yelp diagnostics at RLMRec weight $\alpha = 0.03$ show that increasing patience from five to 20 permits recovery from an early low-score phase. They are validation-only follow-ups and do not replace the main comparisons at $\alpha = 0.01$. This finding cautions that a matched stopping rule need not be equally suitable for different optimization trajectories. We have not repeated all main comparisons under alternative patience values.

Table 1. Test Recall and NDCG at 20, mean ± sample standard deviation over five paired seeds. Batch size is matched within each dataset. Steam model hyperparameters transfer from Amazon; batch size and patience differ.

Dataset	Model	Recall@20	NDCG@20
Amazon	LightGCN	0.1386 ± 0.0012	0.0849 ± 0.0004
Amazon	Full RLMRec	0.1465 ± 0.0023	0.0894 ± 0.0010
Amazon	Item-only RLMRec	0.1462 ± 0.0013	0.0895 ± 0.0006
Amazon	Hybrid framework	0.1536 ± 0.0016	0.0935 ± 0.0007
Yelp	LightGCN	0.1130 ± 0.0026	0.0713 ± 0.0017
Yelp	Full RLMRec	0.1227 ± 0.0015	0.0769 ± 0.0010
Yelp	Item-only RLMRec	0.1225 ± 0.0016	0.0773 ± 0.0005
Yelp	Hybrid framework	0.1199 ± 0.0010	0.0757 ± 0.0006
Steam	LightGCN	0.1319 ± 0.0007	0.0834 ± 0.0005
Steam	Full RLMRec	0.1424 ± 0.0009	0.0903 ± 0.0005
Steam	Item-only RLMRec	0.1410 ± 0.0006	0.0893 ± 0.0002
Steam	Hybrid framework	0.1471 ± 0.0002	0.0931 ± 0.0002

Uncertainty and repeatability. We report sample standard deviations and pointwise paired 95% Student-t intervals over five seeds. Holm adjustment covers twelve contrasts per metric: full and item-only versus LightGCN, and item-only and the hybrid framework versus full, on each of the three datasets. We also report exhaustive two-sided sign-flip values; their minimum with five pairs is 0.0625.

Four repeated executions per RLMRec variant at Amazon seed 42 give Recall@20 SDs of 0.000536 for full and 0.000344 for item-only. This establishes execution variation at that particular setting, including variation when the selected epoch is unchanged. It does not establish a universal detection threshold or quantify noise on Yelp and Steam. Main uncertainty is assessed with the paired seed intervals, not a threshold extrapolated from these repeats.

4 Results

The graph recipe against profile alignment. Table 2 gives the paired differences. The hybrid framework exceeds full RLMRec by 0.00705 Recall@20 on Amazon (4.8% relative) and 0.00474 on Steam (3.3% relative), and falls below it by 0.00281 on Yelp (2.3% relative). The direction holds in all five seeds per dataset. All three paired t comparisons survive the stated Holm adjustment; the exact sign-flip values are 0.0625, so significance depends on the parametric analysis.

The Steam margin is evidence that the Amazon model recipe can transfer to a further benchmark without a new model hyperparameter sweep. It does not remove the unequal development histories: Yelp received validation tuning, Amazon was the development benchmark, and Steam uses a longer stopping budget.

Table 2. Hybrid framework minus full RLMRec, test Recall@20 over five seeds. Pointwise paired t intervals; p_h is Holm-adjusted within the twelve-contrast family, and p_s is the exact two-sided sign-flip value.

Dataset	Difference	95% t interval	p_h	p_s
Amazon	+0.00705	[+0.00295, +0.01115]	0.0354	0.0625
Yelp	−0.00281	[−0.00400, −0.00162]	0.0197	0.0625
Steam	+0.00474	[+0.00376, +0.00572]	0.0017	0.0625

Table 3. Validation Recall@20 on development seed 2023 with alignment switched off ($\alpha = 0$), against LightGCN and item-only RLMRec. Batch size is 1024 on Amazon and 4096 on Yelp and Steam; patience differs as stated in text.

Dataset	LightGCN	Item-only	Graph, $\alpha = 0$	−LightGCN	−Item-only
Amazon	0.13953	0.14991	0.15518	+0.01565	+0.00526
Yelp	0.11899	0.12383	0.10357	−0.01542	−0.02026
Steam	0.13594	0.14201	0.14462	+0.00869	+0.00262

Removing alignment: exploratory evidence. Table 3 reports development-seed validation scores after setting $\alpha = 0$. On Amazon and Steam the recipe exceeds the listed LightGCN and item-only baselines, whereas on Yelp it falls below both. These are single-seed diagnostics, not five-seed test results or comparisons with full RLMRec. They also have unequal patience on Amazon and Yelp: Amazon uses 5/5/25 checks for LightGCN/item-only/graph, Yelp 25/5/5, and Steam 25/25/25. Thus only Steam matches patience across these rows. Other recipe differences remain even when alignment is removed, so the contrasts do not isolate a causal effect of graph structure alone.

Graph-label permutation controls. An isomorphic relabelling keeps the graph topology and degree multiset while changing the correspondence between items and neighbourhoods. Table 4 shows lower scores after this intervention on all three datasets. The Amazon row is a five-seed 64D test comparison with legacy provenance limitations; Yelp and Steam are single-seed 32D validation comparisons. These rows do not form one matched three-dataset experiment.

On Steam, without alignment, the permuted recipe reaches 0.13549 compared with 0.14462 for the real graph and 0.13594 for LightGCN. The real and permuted runs select epoch indices 357 and 324. This supports the relevance of item-to-neighbourhood correspondence in that setting. It does not prove that all of the main-test gain is semantic, rule out optimization effects, or explain the Yelp comparison with RLMRec.

5 Limitations and Conclusion

The Yelp loss is unexplained. We measured the agreement between each item's semantic neighbourhood and its collaborative co-occurrence, expecting it to pre-

Table 4. Isomorphic graph permutation: identical topology and degree multiset, reassigned item labels. Difference is permuted minus real. Settings differ across rows; the Amazon baseline has incomplete historical protocol metadata.

Dataset	Setting	Real graph	Permuted	Difference
Amazon	64D, kd=0.03, 5 seeds, test	0.16676	0.15076	−0.01600
Yelp	32D, kd=0.1, k=20, seed 2023, validation	0.12110	0.11465	−0.00646
Steam	32D, kd=0, seed 2023, validation	0.14462	0.13549	−0.00914

dict where the graph helps, and recorded a prediction for Steam before its development runs. The prediction failed. A subsequent correction to that diagnostic's null distribution produced an ordering consistent with the outcomes, but selecting a measure after seeing three results is not evidence, and we do not present it as a mechanism. The artifact contains the original prediction, its falsification, and the correction.

Our scope is one collaborative backbone, three released benchmarks, fixed offline semantic inputs and Apple MPS execution, for which fixed-seed execution variation was measured at one Amazon setting. Batch size and patience vary across datasets, so cross-dataset contrasts are not isolated effects of dataset identity. The Amazon and Yelp configurations have unequal tuning histories; the Steam model hyperparameters transfer without a new sweep, but the training protocol changes. We do not measure profile-generation cost, and removing user alignment does not remove user interaction data or address cold start, since item profiles may themselves be derived from user-authored text.

The supported conclusion is that the proposed hybrid framework improves five-seed test performance over full RLMRec on Amazon-book and Steam and reduces it on Yelp under the recorded protocols. Alignment-removal and permutation diagnostics support further study of semantic graph correspondence, but their scope and mismatched settings prevent a complete causal account. The failed semantic-agreement prediction leaves the Yelp loss unresolved.

Artifact and provenance. The accompanying artifact includes all 60 main results, their hash-checked manifest, per-seed statistics, graph controls, the recorded Steam prediction, the implementation audit and the scripts that regenerate the tables. It preserves the exact source snapshot for the separate alignment-weight study; complete historical source bytes are not recovered for every other run. Raw inputs are acquired separately from RLMRec, and their upstream generation history is incompletely verified.

AI assistance. AI assistants supported framing, code, statistical checks, experiment execution, literature search and drafting. All reported numbers come from archived training outputs and the artifact's analysis scripts.

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